What is missing for primal-frame approximation?

A Gram-pseudoinverse criterion and a near-Parseval obstruction

Autonomous solve-first investigation, lane 12

13 August 2026

Abstract

Petersen and Raslan prove sparse approximation rates using the canonical dual of a hybrid shearlet–wavelet frame and ask for the corresponding rates using the primal frame. We identify the exact coefficient-level gap. If T is the analysis operator and $G = TT^*$ is the Gramian, then canonical-dual analysis is $G^\dagger T$, where $G^\dagger$ is the Moore–Penrose inverse. Thus weak-ℓ^p boundedness of $G^\dagger$ transfers the proved coefficient sparsity to primal N-term rates. We also prove this cannot be deduced from frame bounds, even near tightness: for every $0 < \varepsilon < 1$ there is a Riesz basis with frame bounds $1-\varepsilon, 1+\varepsilon$ and a vector with one nonzero primal analysis coefficient, yet its best primal N-term error decays only as $(\log N)^{-1/2}$. The remaining hybrid-specific problem is therefore a localization theorem for the pseudoinverse Gramian, not another frame-bound estimate.

1 The source question

Let $\Phi = (\phi_n)_{n\geq 1}$ be a frame for a Hilbert space $\mathcal{H}$, with analysis operator

$$Tu = (\langle u, \phi_n\rangle)_n,$$

synthesis operator T^*, frame operator $S = T^*T$, and canonical dual $\widetilde{\Phi} = (S^{-1}\phi_n)_n$. Petersen–Raslan [1] prove decay of Tu for a class of functions with cartoon-like derivatives. Consequently they obtain fast N-term reconstruction by the *dual* atoms:

$$u_N = \sum_{n\in E_N} \langle u, \phi_n\rangle S^{-1}\phi_n.$$

Their Outlook asks for the analogous best N-term rate with respect to the primal atoms ϕ_n, as required by their adaptive solver program.

For a sequence a, let (a_n^*) be the nonincreasing rearrangement of its moduli and, for $0 < p < 2$, put

$$\|a\|_{p,\infty} = \sup_{n\geq 1} n^{1/p} a_n^*.$$

Write $G = TT^*$ for the Gramian and $G^\dagger$ for its Moore–Penrose inverse.

2 The exact transfer operator

Theorem 1 (Gram-pseudoinverse transfer)**.** *Let Φ have upper frame bound B, let $0 < p < 2$, and let $u \in \mathcal{H}$. Put $c = Tu$. Then the canonical-dual analysis sequence d of u satisfies*

$$d = (\langle u, S^{-1}\phi_n\rangle)_n = G^\dagger c. \tag{1}$$

If $d \in \ell^{p,\infty}$, then

$$\sigma_N(u;\Phi) \leq \sqrt{B}\left(\frac{p}{2-p}\right)^{1/2} \|d\|_{p,\infty} N^{\frac{1}{2}-\frac{1}{p}}, \tag{2}$$

where $\sigma_N(u;\Phi)$ is the best error using at most N primal atoms. Consequently, a uniform implication

$$Tu \in \ell^{p,\infty} \Longrightarrow \sigma_N(u;\Phi) = O(N^{\frac{1}{2}-\frac{1}{p}})$$

follows whenever $G^\dagger$ is bounded on $\operatorname{ran}(T) \cap \ell^{p,\infty}$ in the weak-ℓ^p norm.

Proof. The standard formula for the Moore–Penrose inverse of TT^* is

$$G^\dagger = TS^{-2}T^*. \tag{3}$$

Indeed, it is zero on $\ker T^*$ and, for $c = Tx \in \operatorname{ran} T$,

$$TS^{-2}T^*c = TS^{-2}T^*Tx = TS^{-1}x.$$

Since S is self-adjoint, the last sequence is precisely the analysis of x by $(S^{-1}\phi_n)_n$. This proves (1).

The primal reconstruction is $u = T^*d$. Let E_N index the N largest entries of d. Boundedness of synthesis and the weak-ℓ^p estimate give

$$\begin{aligned}\left\| u - \sum_{n\in E_N} d_n\phi_n \right\| &\le \sqrt{B}\left(\sum_{n>N}(d_n^*)^2\right)^{1/2} \\ &\le \sqrt{B}\,\|d\|_{p,\infty}\left(\sum_{n>N} n^{-2/p}\right)^{1/2} \\ &\le \sqrt{B}\left(\frac{p}{2-p}\right)^{1/2}\|d\|_{p,\infty}N^{\frac{1}{2}-\frac{1}{p}}.\end{aligned}$$

Taking the best primal approximation proves (2). □

The same argument works for Lorentz–Zygmund sequence classes, preserving the logarithmic factors appearing in [1]. Thus the source's precise rate reduces to preservation of its coefficient class by $G^\dagger$.

3 Near-tight frame bounds do not suffice

Theorem 2 (arbitrarily near-Parseval obstruction)**.** *For every $0 < \varepsilon < 1$ there are a Riesz basis Φ for $\ell^2(\mathbb{N})$ with frame bounds $1-\varepsilon$ and $1+\varepsilon$, and a vector u, such that*

$$Tu = e_1,$$

so dual-frame reconstruction is exact with one atom, but

$$\sigma_N(u;\Phi) \ge c_\varepsilon(\log(N+2))^{-1/2} \qquad (N \ge 2). \tag{4}$$

In particular, the primal error is not $O(N^{-s})$ for any $s > 0$.

Proof. Choose a unit vector $w \perp e_1$ whose coordinates, for $n \ge 2$, are

$$w_n = \frac{C}{\sqrt{n}\log(n+1)}.$$

The normalizing constant C exists because $\sum_{n\ge2}(n\log^2(n+1))^{-1} < \infty$. Put

$$K = e_1 \otimes w + w \otimes e_1, \qquad G = I + \varepsilon K, \qquad U = G^{1/2}, \qquad \phi_n = Ue_n.$$

On $\operatorname{span}\{e_1, w\}$, K has eigenvalues $1, -1$, and it vanishes on the orthogonal complement. Hence G is positive with spectrum in $[1-\varepsilon, 1+\varepsilon]$, so (Ue_n) is a Riesz basis with the asserted frame bounds.

Let $u = U^{-1}e_1$, the first canonical-dual atom. Since U is self-adjoint,

$$\langle u, Ue_n\rangle = \langle e_1, e_n\rangle,$$

and therefore $Tu = e_1$. On the other hand, the unique coefficient sequence for expansion by the primal Riesz basis is

$$d = U^{-1}u = G^{-1}e_1 = \frac{1}{1-\varepsilon^2}(e_1 - \varepsilon w). \tag{5}$$

For any sequence a supported on at most N indices, the lower Riesz bound gives

$$\|u - T^* a\| = \|U(d-a)\| \geq \sqrt{1-\varepsilon}\,\|d-a\|_2.$$

The best such a retains the N largest entries of d. By the integral test,

$$\sum_{n>N} |w_n|^2 \asymp \frac{1}{\log(N+2)}.$$

Together with (5), this proves (4). □

This also exhibits the failure of every weak-ℓ^p transfer with $p < 2$: e_1 lies in all those spaces, whereas $G^{-1}e_1$ does not lie in any of them.

4 Consequence for the hybrid-frame program

Theorem 2 shows that controllable frame bounds, even with a ratio arbitrarily close to one, cannot close the source's primal/dual gap. What remains is a structural estimate. Two equivalent-looking routes are:

1. prove that the hybrid Gramian belongs to an inverse-closed localized matrix algebra acting on the relevant weak sequence class, so its pseudoinverse has the same boundedness;
2. prove that the hybrid frame operator and its inverse preserve the cartoon-derivative model class, since the dual atoms are $S^{-1}\phi_n$.

The source's compact-support and Fourier-decay hypotheses provide local correlation estimates, but the packet does not establish the required global mixed shearlet/wavelet pseudoinverse bound. Thus no primal rate for the specific hybrid frame is claimed.

Eight upgrade routes were tested, including localized matrix algebras, explicit dualizable full-plane shearlets, frame-operator invariance, and PDE stiffness-matrix compressibility. Each returns to the same missing off-diagonal inverse estimate. Bounded exact-title and citation searches on 13 August 2026 found no later resolution of the bounded-domain hybrid primal-rate question. Novelty is plausible, not certified.

References

[1] P. Petersen and M. Raslan, *Approximation properties of hybrid shearlet–wavelet frames for Sobolev spaces*, Adv. Comput. Math. 45 (2019), 1581–1606; arXiv:1712.01047.